

SLAM - Ex 6

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This document contains the answers to VAN Exercise 6. The code for this exercise is in [this link](#). The final version used for this submission corresponds to commit `d58d39d`.

1 Relative Pose Constraints

We construct pose-graph measurements from the independent local bundle adjustment windows computed in Exercise 5. Each window spans a start keyframe c_s and an end keyframe c_e , and its optimized poses are expressed in the local coordinate system of that window. The relative-pose measurement is therefore

$$T_{s \rightarrow e} = T_{c_s}^{-1} T_{c_e}.$$

To assign an uncertainty to this measurement, we compute the joint marginal covariance of the two endpoint poses using `gtsam.Marginals`. Writing this covariance in block form as

$$\Sigma_{s,e} = \begin{bmatrix} \Sigma_{ss} & \Sigma_{se} \\ \Sigma_{es} & \Sigma_{ee} \end{bmatrix},$$

the covariance of the end pose conditioned on the start pose is

$$\Sigma_{e|s} = \Sigma_{ee} - \Sigma_{es} \Sigma_{ss}^{-1} \Sigma_{se}.$$

We use $\Sigma_{e|s}$ as the covariance of the relative-pose constraint in the global pose graph. Since the start pose of every local bundle window is anchored by a strong prior, this conditional covariance is close to the marginal covariance of the end pose.

Figure 1 visualizes the optimized locations and marginal covariances of all camera poses in the first bundle adjustment window.

1.1 First Relative Pose Constraint

For the first bundle adjustment window, spanning keyframes c_0 and c_{10} , the estimated relative transformation is

$$T_{0 \rightarrow 10} = \begin{bmatrix} 0.999688 & 0.024557 & -0.004599 & -0.115182 \\ -0.024600 & 0.999653 & -0.009467 & -0.160271 \\ 0.004365 & 0.009577 & 0.999945 & 8.163340 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Its translation component is

$$t_{0 \rightarrow 10} = \begin{bmatrix} -0.115182 \\ -0.160271 \\ 8.163340 \end{bmatrix}$$

The corresponding conditional covariance is

$$\Sigma_{10|0} = 10^{-6} \begin{bmatrix} 0.0696 & -0.0015 & 0.0757 & -0.0627 & 1.2864 & 0.3464 \\ -0.0015 & 0.0537 & -0.0428 & -1.0642 & -0.1828 & 0.7177 \\ 0.0757 & -0.0428 & 0.3468 & 0.6309 & 2.5082 & 0.2776 \\ -0.0627 & -1.0642 & 0.6309 & 29.8617 & 1.0039 & -32.4737 \\ 1.2864 & -0.1828 & 2.5082 & 1.0039 & 33.9532 & 6.9190 \\ 0.3464 & 0.7177 & 0.2776 & -32.4737 & 6.9190 & 123.7906 \end{bmatrix}$$

The lower-right 3×3 block describes the translation uncertainty:

$$\Sigma_{10|0}^{(t)} = \begin{bmatrix} 2.986 \times 10^{-5} & 1.004 \times 10^{-6} & -3.247 \times 10^{-5} \\ 1.004 \times 10^{-6} & 3.395 \times 10^{-5} & 6.919 \times 10^{-6} \\ -3.247 \times 10^{-5} & 6.919 \times 10^{-6} & 1.238 \times 10^{-4} \end{bmatrix}.$$

The eigenvalues of the full covariance matrix are

$$[1.266 \times 10^{-8}, 1.823 \times 10^{-8}, 1.514 \times 10^{-7}, 1.913 \times 10^{-5}, 3.443 \times 10^{-5}, 1.343 \times 10^{-4}].$$

They are all positive, as expected for a valid covariance matrix. The larger eigenvalues correspond mainly to directions with greater translational uncertainty. These results printed in line 59 in `ex6.py` by function `format_relative_constraint_report`.

Repeating this procedure for all local bundle adjustment windows produced 330 relative-pose constraints between consecutive keyframes. Adjacent keyframes are separated by 10 frames.

2 Pose Graph Optimization

Using the relative-pose constraints extracted from the local bundle adjustment windows, we constructed a global pose graph whose variables are the keyframe poses. For every consecutive keyframe pair (c_s, c_e) , we added a `gtsam.BetweenFactorPose3` with measurement $T_{s \rightarrow e}$ and Gaussian noise model defined by the corresponding conditional covariance $\Sigma_{e|s}$. To remove the global gauge freedom, we anchored the first keyframe with a `gtsam.PriorFactorPose3` centered at its pose estimate from Exercise 3, using $\sigma = 10^{-6}$.

2.1 Initialization

We initialized the first keyframe using the prior pose and obtained every later keyframe pose by composing the relative-pose measurements along the chain:

$$\hat{T}_{c_0} = T_{c_0}^{\text{prior}}, \quad \hat{T}_{c_{i+1}} = \hat{T}_{c_i} T_{c_i \rightarrow c_{i+1}}.$$

BA for First Window with Covariance

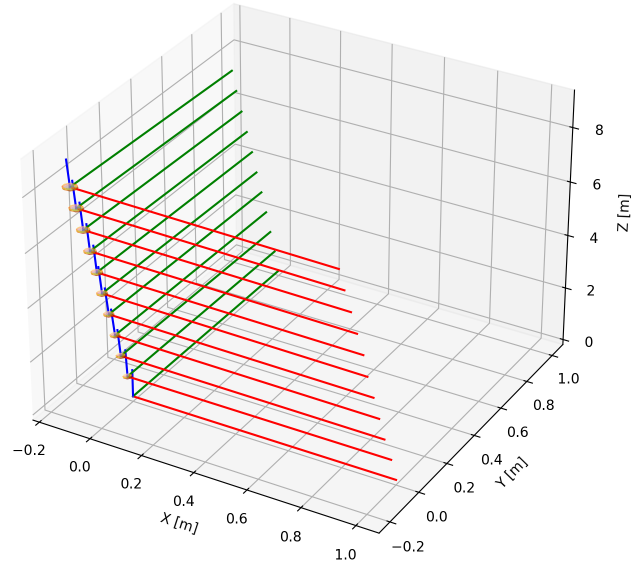


Figure 1: Optimized camera locations in the first bundle adjustment window, including marginal covariance visualizations for all poses in the window. Figure created by the function `plot_bundle_window_trajectory _with_covariances` called in line 64 in `ex6.py`.

This is a natural initialization because each measurement already estimates the motion between consecutive keyframes. Chaining these measurements produces a trajectory that is consistent with the available pose-graph constraints and provides the optimizer with a feasible starting point.

Figure 2 shows the initial keyframe locations supplied to the optimizer.

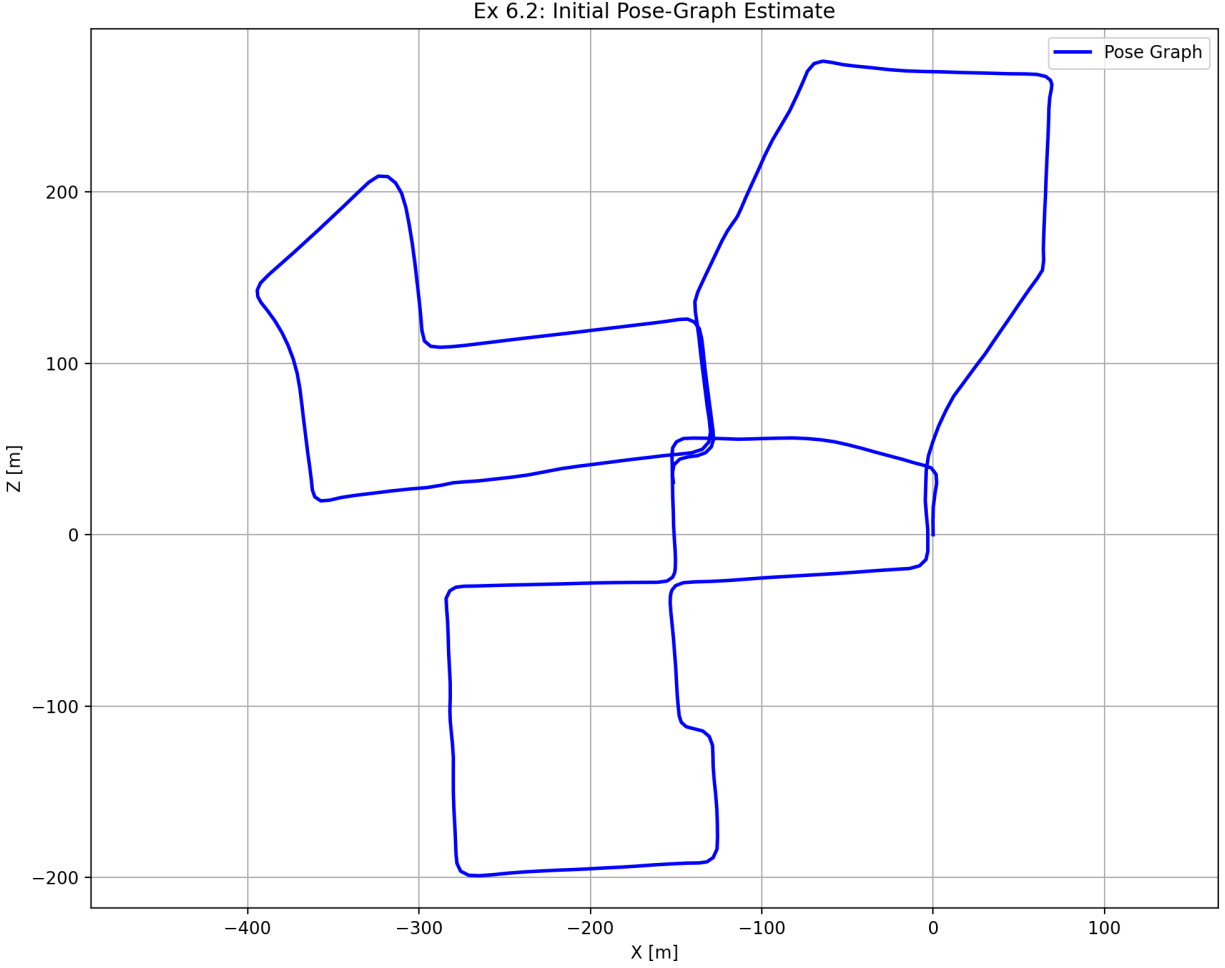


Figure 2: Initial keyframe poses obtained by chaining the relative-pose measurements. Figure created by `plot_pose_graph_top_down()` called from `ex6.py`.

2.2 Optimized Keyframe Locations

We optimized the pose graph using the Levenberg–Marquardt optimizer. The initial and final factor-graph errors are computed in `optimize_pose_graph()` and printed by `format_pose_graph_report`.

Table 1: Optimization statistics for the global pose graph.

Quantity	Value
Initial error	0.00
Final error	0.00

The optimization does not change the keyframe locations. This is expected because the graph contains only consecutive relative-pose constraints and one prior factor; therefore, it is an open chain without loop closures. The initial trajectory is constructed by composing exactly the same relative measurements used in the `BetweenFactorPose3` factors, so ev-

ery factor is already satisfied up to numerical precision. Since there are no redundant or conflicting constraints, the optimizer has no lower-cost configuration to find.

Figure 3 shows the optimized keyframe locations without covariances.

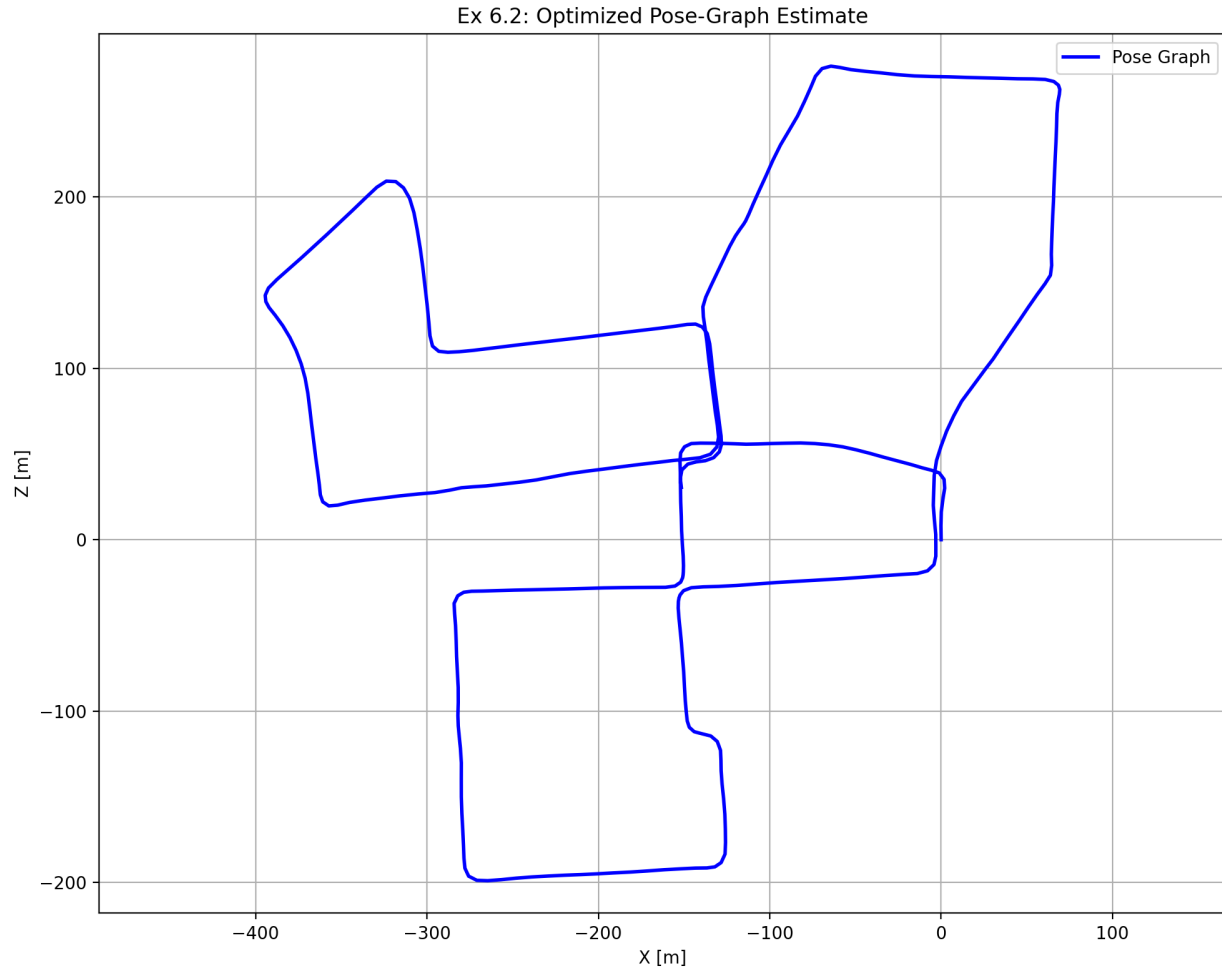


Figure 3: Optimized keyframe locations without covariance visualization. The result is visually identical to the chained initialization. Figure created by `plot_pose_graph_top_down()` called from `ex6.py`.

2.3 Optimized Locations with Marginal Covariances

We computed the final marginal covariances from the optimized pose graph using `compute_pose_graph_marginals()`. Figure 4 shows the optimized trajectory together with projected 3-sigma translation covariance ellipses in the X-Z plane.

Because the first keyframe is strongly anchored, its uncertainty is very small. As relative-pose uncertainty is propagated along the open chain, the marginal uncertainty of later keyframes increases. The growing ellipses therefore reflect the accumulation of uncertainty in global pose estimates when no loop-closure constraints are available.

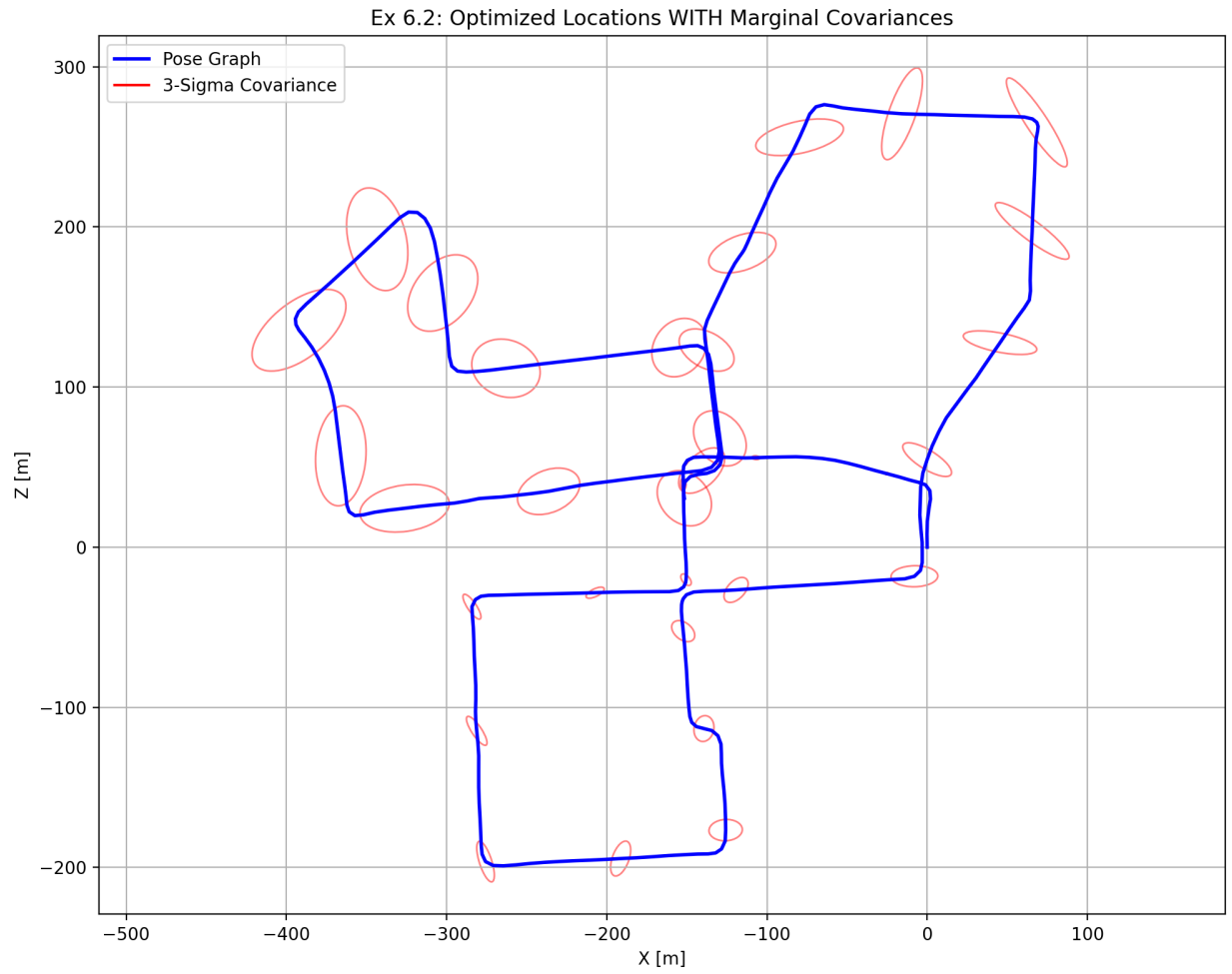


Figure 4: Optimized keyframe locations with projected red 3-sigma translation covariance ellipses in the X-Z plane. Figure created by `plot_pose_graph_top_down()` using marginals computed by `compute_pose_graph_marginals()`, both called from `ex6.py`.